

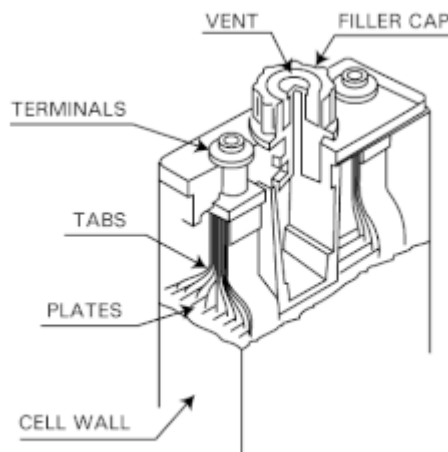
UNIT-II

Energy Storing Devices

LP4: Charging and discharging reactions in a reversible cell (Nickel-Cadmium battery) - Construction, working and applications

Nickel Cadmium battery:

- Nickel-cadmium battery is a source for DC voltage.
- Due to its properties and advantages, it is taking over lead acid-based batteries and gaining popularity in recent times.
- It is small, compact, easily traveled from one place to another. General uses of this battery are toys, calculators, small DC motors, etc.
- Principle wise it is the same as lead accumulator based batteries.
- A metal is rolled with cadmium and separator layers and kept in redox so that the chemical reaction produces the DC voltage.
- In a nickel-cadmium battery, the redox material is used as a base, and around it, the layer of nickel and a separator are used.
- The nickel-cadmium cell voltage is around 1.2 V. When connected in series generally 3 to 4 cells are packed together to get an output of 3.6 to 4.8 V

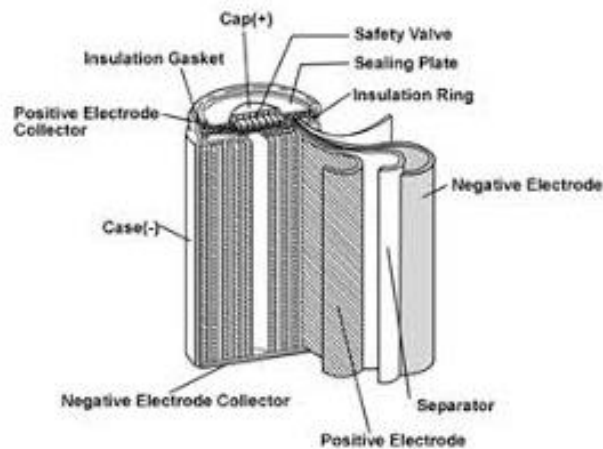


Nickel-Cadmium Battery Design

Nickel-Cadmium Battery Theory

- The operating principle of a nickel-cadmium battery is the same as other batteries.
- To improve efficiency, nickel and cadmium are used.

- Nickel-cadmium battery is the source of DC voltage, hence it must consist of two potential points i.e positive and negative or also called anode and cathode.
- In a nickel-cadmium battery, first, a layer of nickel oxide NiO_2 is kept around the redox.
- This layer of nickel oxide acts as a cathode layer.
- Above the nickel oxide layer, a layer of KOH is kept, which acts as a separator. It must be noted that this separator layer must be soaked in water or moist.
- Its purpose is to provide required OH negative ions, for the chemical reaction. Above the separator layer, cadmium is placed.
- The cadmium layer acts as the anode for the nickel-cadmium battery. The nickel-cadmium battery diagram is shown below.



Nickel-Cadmium Battery Diagram

- As shown, in the diagram, the nickel acts as a positive electrode collector and the cadmium layer acts as a negative layer collector.
- The separator layer between the two layers is made up of KOH or NaOH.
- Its purpose is to provide OH ions. Apart from these, it also consists of a safety valve, sealing plate, insulation ring, insulation gasket, and an outer case.
- The purpose of the insulator ring is to provide insulation between the two layers.
- The insulator gasket is the place where the insulation ring is kept nearby. The separator layer is connected to this ring.
- The outer case is to provide protection to the inner layers from external factors such as damages and mishandling of the battery.

- It must be noted that, due to chemical reactions taking place within the battery, it always hazardous to work with the battery.

Nickel Cadmium Battery Equations

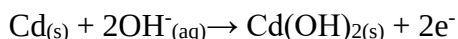
Nickel Cadmium battery is a secondary battery and it can be recharged by passing electricity. It can operate both as an electrochemical cell and as an electrolytic cell. When it acts as a voltaic cell, it converts chemical energy into electrical energy and vice versa when operates as electrolytic cell.

In this battery cadmium acts as anode and a metal grid containing a paste of NiO₂ acts as a cathode. KOH is used as electrolyte.

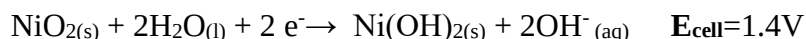
Cell representation: Cd/Cd(OH)₂ // KOH_(aq)/NiO₂/Ni

Discharging (electrochemical cell):

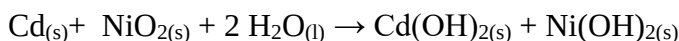
At anode cadmium is oxidized to Cd²⁺ ion which combines with OH⁻ ions to form Cd(OH)₂.



At cathode NiO₂ is reduced to Ni²⁺ ions which also combines with OH⁻ ions to form Ni(OH)₂.



Overall reaction



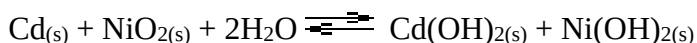
Recharging (electrolytic cell): By passing electricity.

At cathode: Cd(OH)_{2(s)} + 2e⁻ → Cd_(s) + 2OH⁻

At Anode: Ni(OH)_{2(s)} + 2OH⁻ → NiO_{2(s)} + 2H₂O_(l) + 2e⁻

Overall reaction: Cd(OH)_{2(s)} + Ni(OH)_{2(s)} → Cd_(s) + NiO_{2(s)} + 2H₂O_(l)

Net reaction during discharging & recharging is



Nickel-Cadmium Battery Temperature Range

The temperature range for nickel battery is 0 to 45-degree centigrade during charging and -20 to 65 degrees centigrade during discharging. Beyond this temperature range, the battery fails to operate and even chances of explosion exist.

Nickel-Cadmium Battery Toxicity

Nickel-cadmium battery is very toxic to the human body. Cadmium is a heavy metal posing several risks to the human body. Cadmium even has a physiological effect on the system. The average presence of cadmium in the human body is approximately 1 microgram per liter. It has a

direct effect on the digestive system. Similarly, nickel is also poisonous to the human respiratory system.

Nickel-cadmium battery voltage

In general, each voltage for a Nickel-cadmium battery would be approximately 1.2 V. Number of cells are connected in series or parallel to get the required voltage. Apart from the voltage, its specific energy is around 50-60 Wh per Kg. This is moderately high than nickel-iron, but relatively less than nickel-zinc and nickel-metal hydride batteries.

The specific power is 200 W per kg. This is moderately high than nickel-iron but relatively less than nickel-zinc and nickel-metal hydride batteries. For nickel-metal batteries, it is around 170-1000. For nickel-iron batteries, it is around 100. The energy efficiency is around 70-75%. This is moderately high than nickel-iron but relatively less than nickel-zinc and nickel-metal hydride batteries. For nickel-metal batteries, it is around 70-80 %. For nickel-iron batteries, it is around 60-70%.

Construction of Nickel-Cadmium Battery

- Constructional, the nickel-cadmium battery is the same as lead acid-based batteries. It consists of three fundamental layers.
- The first one is a nickel layer, then the separator layer, and the cadmium layer.
- The nickel acts as a positive electrode collector and the cadmium layer acts as a negative layer collector.
- The separator layer between the two layers is made up of KOH or NaOH.
- Its purpose is to provide OH ions. Apart from these, it also consists of a safety valve, sealing plate, insulation ring, insulation gasket, and an outer case.
- The purpose of the insulator ring is to provide insulation between the two layers.
- The insulator gasket is the place where the insulation ring is kept nearby. The separator layer is connected to this ring.
- The outer case is to provide protection to the inner layers from external factors such as damages and mishandling of the battery.
- Due to chemical reactions taking place within the battery, it is always hazardous to work with the battery.

- The layers along with the separator layer form the required chemical reaction and form the potential difference.

Advantage

- It is higher energy efficiency, smaller and lighter.
- Has longer life than lead acid.
- Like dry cell, it can be packed in a sealed container.

Disadvantage

It is more expensive than lead acid storage battery.

Uses

- It is used in calculators, electronic flash units, transistors, electrical shavers, emergency light, toys and medical instruments.
- It is used in aircraft and space satellite power system